# FLUID DYNAMICS

KATTA NAVEEN

27-07-2024

# INTRODUCTION

- The phenomena considered in the fluid dynamics are macroscopic, So a fluid is regarded as the continuous medium.

- If we are having fluid velocity (v), Pressure (P), Density (p) the state of the fluid is fully determined.

- All these quantities are, in general function of the coordinates x,y,z and of time (t). Where (x,y,z) refers to fixed points in space and not a specific particles of the fluid.

# CONTINUITY EQUATION

**Conservation of Mass :**

**The total Mass of the liquid flowing out of the volume V in unit time is,**

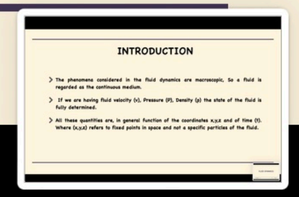
$$\oint \rho \, \mathrm{v} \cdot \mathrm{df}$$

**The Decrease per unit time in the mass of the fluid in the volume V can be written as ,**

$$-\frac{\partial}{\partial t} \int \rho \, \mathrm{d}V$$

**Equating these two we get :**

$$\int [\frac{\partial \rho}{\partial t} + \nabla(\rho v)] \, dV \; = \; 0$$



# EULER'S EQUATION

The total force acting on the volume is equal to the integral :

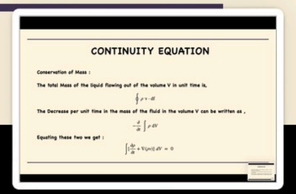
$$-\oint p\mathrm{df} = -\int \mathrm{grad}\, p \ \mathrm{d}V$$

From this force acting on the volume dV will be -grad p . Then we can write $\rho \mathrm{d}v/\mathrm{d}t = -\,\mathrm{grad}\, p\,.$

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\cdot\mathrm{grad})\mathbf{v}\,.$$

From the above equations we write the equation of motion of the fluid.

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\cdot\mathrm{grad})\mathbf{v} = -\frac{1}{\rho}\,\mathrm{grad}\, p$$



For a fluid at rest in an gravitational field the Euler's equation will become, $\operatorname{grad} p = \rho \mathbf{g}$.

If the density is constant throughout the volume there will be no compression of the fluid under the action of external force then the above equation becomes :

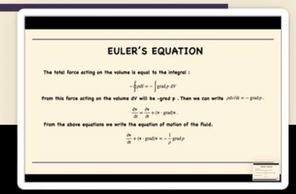
$$\partial p/\partial x = \partial p/\partial y = 0, \quad \partial p/\partial z = -\rho g, p = -\rho g z + \text{ constant.}$$

Now we derive the equation of equilibrium for a very large mass of fluid, whose separate parts are held together by gravitational attraction , a "star".

We know that $\triangle \phi = 4\pi G\rho$ and the condition for equilibrium is $\operatorname{grad} p = -\rho \operatorname{grad} \phi$.

From the above equation see get $\operatorname{div}\left(\frac{1}{\rho}\operatorname{grad} p\right) = -4\pi G\rho$

If the body is not rotating, it will be spherical when not in equilibrium and density and pressure distributions will be in spherical equilibrium then the equations becomes , $\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{r^2}{\rho}\frac{\mathrm{d}p}{\mathrm{d}r}\right) = -4\pi G\rho$



# BERONOULLI'S EQUATION

**STEADY FLOW means one in which the velocity is constant in time at any point occupied by the fluid then we have the equation ,** $\frac{1}{2}\operatorname{grad} v^2 - v \times \operatorname{curl} v = -\operatorname{grad} w$.

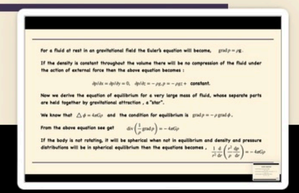
**The tangent to the STREAMLINE of any point gives the direction of the velocity at that point.**

$$\frac{dx}{v_x} = \frac{dy}{v_y} = \frac{dz}{v_z}.$$

**From the two equations we get** $\frac{\partial}{\partial l}\left(\frac{1}{2}v^2 + w\right) = 0$ .

$\frac{1}{2}v^2 + w =$ **constant. Where l is along the streamline .This is Bernoullis equation.**

**If we take the fluid in Gravitational field** $\frac{1}{2}v^2 + w + gz =$ **constant.**



# INCOMPRESSIBLE FLUIDS

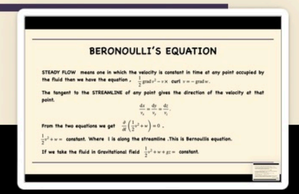
If the density is constant throughout the volume of the fluid and its motion then we speak of incompressible flow.

Then the Eulers equation will be , $\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \operatorname{grad})\mathbf{v} = -\operatorname{grad}\left(\frac{p}{\rho}\right) + \mathbf{g}$

Continuity equation will be " div v = 0 " .

We know the equation $\partial \mathbf{v}/\partial t + (\mathbf{v} \cdot \operatorname{grad})\mathbf{v} = -\operatorname{grad} w$. from this and the first equation we can write "w"(heat function) as "P/p" .

Therfore the Bernoullis equation will simply become $\frac{1}{2}v^2 + p/\rho + gz =$ constant .



# GRAVITY WAVES

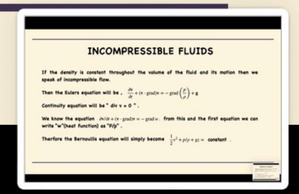
If under the action of some external perturbation the free surface of a liquid in equilibrium is moved motion will occur in the fluid . This motion will be propagated over the surface in the form of waves with are Called gravity waves.

After considering velocity to be small and incompressible liquid we get $p = -\rho g z - \rho \partial \phi / \partial t$ .

By taking some vertical displacement I.e, over z $(\partial \phi / \partial z)_{z=\zeta} = \partial \zeta / \partial t = -\left( \frac{1}{g} \frac{\partial^2 \phi}{\partial t^2} \right)_{z=\zeta}$ .

Since the oscillations are small we can take derivative at z=0 .

$$\triangle \phi = 0; \left( \frac{\partial \phi}{\partial z} + \frac{1}{g} \frac{\partial^2 \phi}{\partial t^2} \right)_{z=0} = 0.$$


INCOMPRESSIBLE FLUIDS

If the density is constant throughout the volume of the fluid and its motion then we speak of incompressible flow.

Then the Eulers equation will be , $\frac{\partial v}{\partial t} + (v \cdot \text{grad})v = -\text{grad}\left(\frac{p}{\rho}\right) + g$

Continuity equation will be " div v = 0 " .

We know the equation $\partial v/\partial t + (v \cdot \text{grad})v = -\text{grad}\,w$ from this and the first equation we can write "w"(heat function) as "Hy" .

Therfore the Bernoullis equation will simply become $\frac{1}{2}v^2 + gy/\rho + p/\rho = \text{constant}$

**CONDITIONS :**

**waves on the surface of the liquid whose area is unlimited.**

**We shall also consider the wavelength is small in comparison with the depth of the liquid.**

**If the gravity wave is propagated along the x-direction it will be simply** $\phi = f(z)\cos(kx - \omega t)$ .

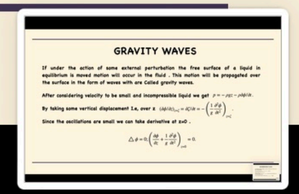
**After solving these equation see get**

$$x - x_0 = -A\frac{k}{\omega}e^{kz_0}\cos\left(kx_0 - \omega t\right)$$

$$z - z_0 = -A\frac{k}{\omega}e^{kz_0}\sin\left(kx_0 - \omega t\right)$$

**The velocity of propagation of gravity waves in**

$$U = \frac{1}{2}\sqrt{(g/k)} = \frac{1}{2}\sqrt{(g\lambda/2\pi)}$$



# LONG GRAVITY WAVES

Here we consider the wave length is large compared with the depth of liquid so we examine the propagation of wave in a Chanel. Where the cross section is S=S(x,t) and we write equation of continuity $\frac{\partial S}{\partial t}+\frac{\partial (Sv)}{\partial x}=0.$

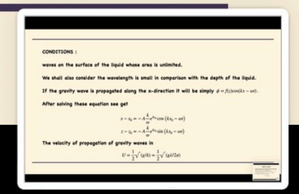
The velocity along x-direction will be large compared to y-direction and z-direction.

The x-component of Euler's equation can then be written in the form $\frac{\partial v}{\partial t}=-\frac{1}{\rho}\frac{\partial p}{\partial x}$

The z component in the form $\frac{1}{\rho}\frac{\partial p}{\partial z}=-g$

From both equations we obtain $\partial v/\partial t=-g\partial\zeta/\partial x\,.$ considering z=e is the free surface.

After solving these equations we get velocity of the wave propagated is $U=\sqrt{\left(gh_0\right)}\,.$


CONDITIONS :

waves on the surface of the liquid whose area is unlimited.

We shall also consider the wavelength is small in comparison with the depth of the liquid.

If the gravity wave is propagated along the x-direction it will be simply $\phi = f(t)\cos(kx - \omega t)$.

After solving these equation see get

$$x - z_0 = -\frac{k}{\omega}\int v\sin(kz_0 - \omega t)$$
$$z - z_0 = -\frac{k}{\omega}\int v\sin(kz_0 - \omega t)$$

The velocity of propagation of gravity waves is

$$U = \sqrt{\frac{\lambda}{2\pi}}\sqrt{g\tanh\frac{2\pi}{\lambda}h} = \sqrt{\frac{\lambda}{2\pi}}\sqrt{g\tanh kh}$$

# INTERNAL WAVES

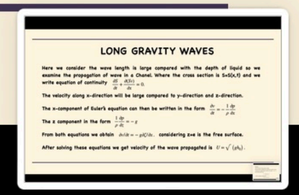
These are a kind of gravity waves which can be propagated in incompressible fluids. Such waves are due to inhomogenity caused by gravitational field.

The pressure necessarily varies with height hence any displacement of fluid particle in height destroys the mechanical equilibrium and causes oscillatory motion.

We can neglect the density change caused due to pressure.the change in density caused by the thermal expansion cannot be neglected.

Firstly we write the equation of conservation of entropy to first order of smallness.

$$\partial s'/\partial t + \mathbf{v} \cdot \operatorname{grad} s_0 = 0$$



# INTERNAL WAVES

We can neglect the (v.grad)v in euler's equation since we are considering small osscillations

$$\frac{\partial \mathbf{v}}{\partial t} = -\frac{\operatorname{grad} p}{\rho} + \mathbf{g} = -\frac{\operatorname{grad} p'}{\rho_0} + \frac{\operatorname{grad} p_0}{\rho_0^2}\rho'$$

The euler equation will become

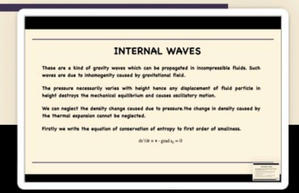
$$\frac{\partial \mathbf{v}}{\partial t} = \frac{\mathbf{g}}{\rho_0}\left(\frac{\partial \rho_0}{\partial s_0}\right)_p s' - \operatorname{grad}\frac{p'}{\rho_0}.$$

We seek solution to be of the form

$$\mathbf{v} = \text{constant} \times e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$$

We find velocity to be

$$U = \left(\omega_0/k\right)\cos\theta \qquad \mathbf{U}\cdot\boldsymbol{v} = -\left(\omega_0/k\right)\cos\theta\sin\theta$$


INTERNAL WAVES

These are a kind of gravity waves which can be propagated in incompressible fluids. Such waves are due to inhomogenity caused by gravitational field.

The pressure necessarily varies with height hence any displacement of fluid particle in height destroys the mechanical equilibrium and causes oscillatory motion.

We can neglect the density change caused due to pressure,the change in density caused by the thermal expansion cannot be neglected.

Firstly we write the equation of conservation of entropy to first order of smallness.

$\partial s'/\partial t + \mathbf{v}\cdot\operatorname{grad} s_0 = 0$

# THANK YOU